

Math 335 HW1

Problems:

1) Determine the order of the DE and whether it is linear or nonlinear.

(a) $y' + \sin y = 0$

(b) $y' + e^t = 0$

(c) $y''' + t^2 y'' - 5y' + \sin(t)y = f(t)$

(d) $y''' + 5y^4 - \cos t = 0$

(e) $\frac{y'}{y} = 1 + t$

(f) $\frac{y'}{y} = 1 + y$

2) Plot the slope field of the DE. If $y(0) = 0$ determine the behavior of the solution as t approaches infinity.

(a) $y' = y^2 - 1$

(b) $y' = t^2 - 1$

3) Solve the IVP.

$$y' = \frac{1}{t^2 - 3t + 2}, \quad y(0) = 1$$

4) Verify that y_1 is a solution to the IVP.

$$t^2 y'' + 5t y' + 4y = 0, \quad y(1) = 0$$
$$y_1 = t^{-2} \ln t$$

5) Solve the DE.

$$y' + 2y = \frac{e^{-2t}}{1 + t^2}$$

6) Solve the DE.

$$y' - \frac{1}{t} \sin t^2 = -\frac{2}{t} y$$

7) Solve the DE.

$$y' + \cot(t) y = e^t \cot t$$